The Ideas Page: Crackdown on Jantar Mantar 2.0, amid other closures, calls for redemocratisation 9

TUESDAY, OCTOBER 13, 2026

The Indian EXPRESS

JOURNALISM of COURAGE

NEW DELHI, LATE CITY, 18 PAGES

₹7.00 (₹8 RAIPUR, ₹15 SRINAGAR) • www.indianexpress.com

DAILY FROM: AHMEDABAD, CHANDIGARH, DELHI, JAIPUR, KOLKATA, LUCKNOW, MUMBAI, NAGPUR, PATNA, PUNE, VADODARA

DAYS AFTER RATE HIKE, PRESSURE ON RBI

Rising food prices push retail inflation to 5.54% in Sept

Siddharth Upasani
New Delhi, October 12

DAYS AFTER the Reserve Bank of India (RBI) increased interest rates for the first time in over three-and-a-half years, government data released on Monday showed that headline inflation rate surged from 4.83% in August to 5.54% in September, driven mostly by high food prices.

While there was a month-on-month increase in overall prices, an unfavourable base effect also played a role in pushing up the headline inflation rate.

In a sign that price pressures are spreading, core inflation rose from 4.1% in August to 4.3% in September. Core inflation measures how the price of the household consumption basket has changed, after excluding food and fuel items whose prices can move sharply, tending to have an outsized influence on the overall inflation rate. As such, economists like to narrow down on the core inflation rate to understand underlying price pressures.

According to data released by the Ministry of Statistics and Programme Implementation (MoSPI) on Monday, food inflation rose sharply from 5.97% in August to 7.74% in September,

»CONTINUED ON PAGE 2



Beyond anti-submarine warfare, INS Mangrol can conduct underwater surveillance, low-intensity maritime operations, and mine warfare tasks. EXPRESS

Navy inducts INS Mangrol, to help detect and counter underwater threats

Manish Kumar Pathak
Mumbai, October 12

INS Mangrol, the third vessel in the Mahe-class Anti-Submarine Warfare Shallow Water Craft (ASW-SWC) series, was commissioned at the Naval Dockyard in Mumbai Monday, strengthening the Indian Navy's ability to detect and counter enemy submarines in shallow coastal waters.

Designed to operate in shallow and confined waters where larger warships may face limitations, the indigenously built vessel will help protect ports, harbours, and coastal approaches against underwater threats.

Built by Cochin Shipyard Limited (CSL), Kochi, INS Mangrol is equipped with advanced sonar, lightweight torpedoes and other anti-submarine systems that enable it to detect, track, and counter hostile submarines. With more than 80 per cent indigenous content, the vessel also reflects the Navy's push to strengthen domestic capabilities in warship construction and defence technology.

Vice Admiral Sanjay Vatsayan, Flag Officer Commanding-in-Chief of the Western Naval Command, commissioned the ship.

During the commissioning ceremony, Vice Admiral Vatsayan said the Navy had commissioned 10 ships in 2026, with another eight expected to enter service by the end of the year.

"Our fleet currently comprises 151 platforms, including 131 ships and 20 submarines," he said.

"We currently have 42 ships under construction at Indian shipyards, while Acceptance of Necessity (AoN) has been accorded for another 69 ships, six submarines and 120 Fast Interceptor Craft. With these additions, our fleet is projected to grow to more than 170 platforms over the next few years, with a new ship being commissioned every 40 to 50 days. By 2034, approximately 60 per cent of our platforms are expected to be less than 10 years old on average," Vatsayan said.

The induction of the Mahe-class vessels is part of the Navy's efforts to bolster its coastal anti-submarine warfare capabilities and replace the ageing Abhay-class corvettes, which entered service between 1989 and 1991.

Their shallow-water design is intended to address the specific challenges of detecting and responding to submarines operating close to the coast, including around strategically important ports and maritime infrastructure.

INS Mangrol is equipped with the Abhay hull-mounted sonar, lightweight torpedoes, an indigenous anti-submarine rocket launcher and the Mareech Advanced Torpedo Decoy System. Together, these systems help the ship detect and track underwater threats, engage hostile submarines, and deploy countermeasures against incoming torpedoes.

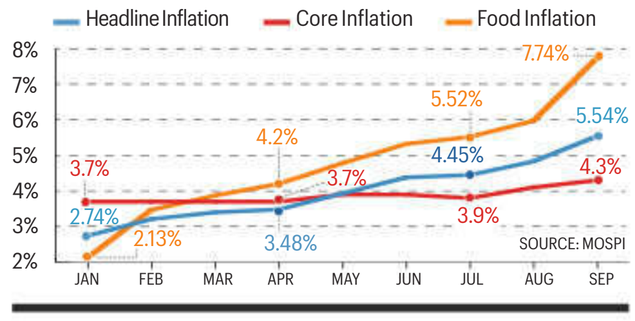
Powered by three diesel engines and fitted with waterjet propulsion, the ship has high manoeuvrability, making it suitable for operations in shallow and confined waters. Beyond anti-submarine warfare, it can conduct underwater surveillance, low-intensity maritime operations, and mine warfare tasks.

It can also work alongside maritime patrol aircraft, helicopters and other naval vessels to strengthen surveillance and anti-submarine operations.

INS Mangrol has been built under the Centre's Make in India and Aatmanirbhar Bharat initiatives. It is named after the coastal town of Mangrol in Gujarat's Junagadh district.

Retail inflation

• Household inflation moves closer to 6%



driven by higher prices of vegetables, fruits and sugar-related items. According to ICRA Chief Economist Aditi Nayar, of the 71-basis-point increase in inflation in September, 62 bps was because of the rise in prices of items from the 'food and beverages' category.

Food apart, prices of consumer electronics also kept rising, reflecting the shortage in memory chips, a key input in manufacturing these goods.

The latest inflation print of 5.54% is the highest in the new series which has data starting from January this year. If the old series, which had 2012 as the base year, is also taken into account, then the September inflation rate is the highest in almost two years, with the previous high being 6.21% in October 2024.

Economists now see headline inflation crossing 6% in October, data for which will be released on November 12.

The RBI is legally mandated to keep the headline inflation rate at 4% in a band of 2-6%. However, this is the fourth month in a row that inflation has come in above 4%. The central bank, which last week raised the policy repo rate by 25 basis points (bps) to 5.5%, expects CPI inflation to average 4.9% in July-September, 6% in October-December, 5.7% in January-March 2027, and 5.6% in April-June 2027.

While the average inflation for July-September is 4.9%, in line with the RBI's forecast, the "challenge" will be in October-December, said Madan Sabnavis, Chief Economist at Bank of Baroda. "Our view still holds that December will warrant another rate hike," he said.

A key area of concern for policymakers will be how more parts of the household consumption basket are seeing higher prices and increase in inflation, called "generalisation of inflation" by economists. In September, 11 of the 12 major categories of the CPI saw higher inflation, which Devendra Pant, Chief Economist at India Ratings & Research, said is indicative of this phenomenon.

A sub-category that captures the spread of higher prices in the wake of the West Asia war is 'restaurants and cafés'. From 2.73% in February, the consumer inflation reported by 'restaurants and cafés' has risen to 9.39% in September, reflecting higher cooking gas prices and other inputs.

While price index of the overall CPI basket was 0.7% higher in September as compared to August, the index for 'restaurants and cafés' increased by 1% month-on-month. Month-on-month, or sequential, increases are indicative of momentum in prices and watched closely by policymakers.

Sabnavis of Bank of Baroda warned that producers are yet to fully pass on higher costs to consumers. "Industries like durables, FMCG, automobile have announced an increase in prices and should get reflected in the CPI over the next few months. The state of kharif crop is important as food inflation will be largely driven by them. As of now, higher inflation is mainly due to depletion of earlier stocks as we head to the new season. Crop arrivals are hence important."

EXPRESS special

China looming, Arunachal hydel project under fresh protest shadow

Security beefed up, stringent law invoked to arrest protesters, as tribal community fears for loss of livelihood, way of life

Jay Mazoomdaar
New Delhi, October 12

NEARLY A month after columns of security personnel moved into the Ugeng-Komkar area of the Siang valley in Arunachal Pradesh, to facilitate surveys and drilling for the prefeasibility report of the proposed 11,200-MW Siang Upper Multipurpose Project, protests against it are refusing to die down.

Last week, four protesters were detained under the Arunachal Pradesh Unlawful Activities (Prevention) Act, 2014, triggering a fresh round of demonstrations. People hit the streets in Geku, Upper Siang, demanding the release of the four, with the Youth Wing of the Siang Indigenous Farmers' Forum (SIFF) urging that government consultations on the project be boycotted.

For the indigenous Adi community that is anxious about displacement and livelihoods, as well as the loss of land that they link back to their ancestors, the deployment of heavy security and the arrests have only stoked fears.

In late September, Chief Minister Pema Khandu's visit saw an attack on a local MLA's residence by protesters, who also torched a police vehicle. Fresh clashes erupted on October 4, amid allegations of arson by protesters and excessive force by security personnel.

In a sign that the Centre was watching, on October 8, Union Minister Kiren Rijiju, who belongs to Arunachal, called for dialogue with senior members of the Adi community. "If we sit down and talk, technically, everything will be clear. We have to talk. We have to move forward," he said. However, the impasse has a long history.

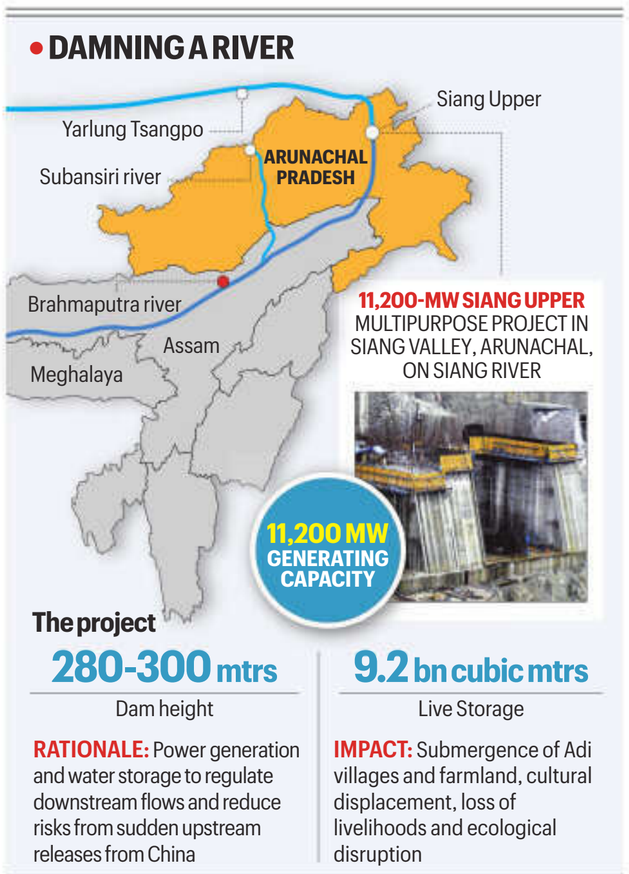
The ferocious Siang barrels through the Himalayas until it approaches the Assam plains, where it becomes the braided Brahmaputra. Its political journey, however, is marked by diversions.

In August 2010, when the Congress was in power in Arunachal, the Itanagar residence of then chairman of the Siang Peoples' Forum and firebrand anti-dam activist Ojing Tasing was raided by police. Following this, Tasing met the then UPA government's Environment Minister Jairam Ramesh in New Delhi, demanding that hydropower projects in the Siang basin be scrapped.

Soon after, the UPA government held a public consultation in Guwahati and Ramesh wrote to then Prime Minister Manmohan Singh, acknowledging that his arguments about the projects' strategic significance had not convinced everyone. "There were some NGOs from Arunachal Pradesh who... said we should not make Arunachal Pradesh a pawn in the race between India and China," he wrote.

Resistance to big dams elsewhere in Arunachal also found a voice in the opposite camp, in Tarun Vijay, former editor of the RSS mouthpiece *Panchajanya* and then a BJP MP. In May 2012, Vijay told the Rajya Sabha: "Government ministers tell me: 'Tarun ji, do not concern yourself with this; perhaps people from China are instigating them.' Is that a responsible thing to say? If people from China are indeed instigating an agitation... the Government has an even greater responsibility to send a Union minister to speak to those people."

Campaigning in Pasighat in February 2014, before he became the Prime Minister, Narendra Modi also acknowledged the local opposition to large dams and spoke of harnessing hydropower through smaller projects while protecting the environment.



The tables turn

In a memorandum submitted this July, Arunachal Congress president Bosiram Siram argued that concerns over China's upstream activities should not override free and informed consent of local tribal communities. In September, Indian Youth Congress national general secretary Tarh Johny described the approach of the BJP government as "dictatorship", calling for transparency and meaningful consultation with the affected people.

Now leading the charge for the Siang Upper project is Chief Minister Khandu. On September 21, he said he had instructed the Chief Secretary to act against government employees opposing the project.

The biggest U-turn is by Tasing, now among Khandu's most vocal lieutenants and Arunachal Rural Development and Panchayati Raj Minister. The activist now says: "If we don't construct our dam, what will be the impact on us if China uses its dam as a water bomb on India?"

Attempts to approach him over his changed position were unsuccessful.

The argument

Underlining the political reversals is a strategic argument that remains consistent: India must develop its downstream water resources as China expands its hydel capacity on the Yarlung Tsangpo, as the Siang is known in upstream Tibet.

Initially, the China factor was not dominant when India proposed to harness the Siang and Subansiri rivers back in 1982. As Chinese plans on Yarlung Tsangpo emerged, New Delhi and Beijing agreed on a mechanism in 2002 for China to provide flood-season hydrological data.

In 2003, India launched a nationwide initiative to develop 50,000 MW of hydropower, with around 27,000 MW envisaged from Arunachal Pradesh, setting off a flurry of agreements with developers in Itanagar.

In 2011, the government told Parliament about a Chinese run-of-the-river project upstream of Siang and stressed the need for Arunachal Pradesh and Assam to harness the Brahmaputra's waters. In 2013, India raised concerns over additional Chinese projects on the Yarlung Tsangpo and asserted its interests as a downstream user.

In 2017, with BJP governments at the Centre and in Arunachal, NITI Aayog proposed replacing the Siang Upper Stage-I and Stage-II projects with a single large storage project. The proposal has since evolved into the 11,200-MW project now at the centre of the protests.

China's plans have also advanced. Beijing approved its 60,000-MW lower Yarlung Tsangpo hydropower project in December 2024. Comprising five cascade hydropower stations, the project seeks to harness the river's steep descent near the Indian border — a drop of nearly 2,000 metres over a 50-km stretch. This has sharpened concerns in India about downstream flows and sudden releases of water, reinforcing the case for storage on the Siang.

But not everybody is convinced. "There is no realistic prospect of establishing the so-called 'first user right' in international jurisprudence. Instead, India should demand project information and possibly assert our right to joint impact assessment to ascertain the risks posed by the dams," says Himanshu Thakkar, coordinator of the South Asia Network on Dams, Rivers and People.

WEB EXCLUSIVE
Countering Houthis should not require turning Mecca's sanctity into a political cause — *Bhopinder Singh*
Log on to www.indianexpress.com

WORDLY WISE
When tillage begins, other arts follow.
—Daniel Webster

The Indian EXPRESS
~ FOUNDED BY ~
RAMNATH GOENKA
IN 1932

BECAUSE THE TRUTH
INVOLVES US ALL

Unregulated use of GM: A cautionary tale

A QUARTER or more of India's cultivated area under cotton comprises genetically modified (GM) hybrids incorporating "unapproved" herbicide-tolerant (HT) and *Bacillus thuringiensis* (Bt) traits, according to a seed industry-commissioned study. This isn't surprising. Cotton farmers today face problems of labour scarcity for manual weeding and pink bollworm insect attacks. They would be inclined to plant hybrids that enable them to spray chemical herbicides and provide protection against a broader spectrum of bollworms and sucking pests. The existing "approved" GM Bt hybrids do not contain genes conferring tolerance to weed killers such as glyphosate or control against pink bollworm. The study shows the presence of unapproved HT and next-generation Bt genes in over half of the seed/leaf samples collected from farmers' fields across 16 districts of Maharashtra, Telangana, Andhra Pradesh and Rajasthan.

The study's findings are a clear demonstration of farmer acceptance of HT technology. For cotton growers, this isn't about ideology. Manual weeding costs, even when labour is available, come to Rs 12,000-15,000 per acre now, whereas it is only Rs 750-850 for 2-3 rounds of glyphosate application. The same goes for hybrids harbouring the Vip3A Bt gene. While derived from the same soil bacterium as the Cry1Ac and Cry2Ab genes present in the "approved" Bt hybrids, it has a different mode of action that is effective even against the pink bollworm. Neither the glyphosate tolerance nor the Vip3A Bt gene traits have, however, been officially authorised for insertion into cotton plants for commercial cultivation. Farmers are, then, growing their hybrids via sourcing of seeds from illegal channels. That includes traders and breeders who have undertaken introgression of the traits, bypassing the government's biosafety and regulatory approval system.

The loser is not just the organised seed industry. Sales of "legal" Bt cotton hybrid seeds have fallen from 59.6 million to 46.3 million packets between 2022-23 and 2025-26. This, even as the unauthorised HT-Bt seed market has grown to an estimated 8.2-8.8 million packets. A bigger loser is the farmer, who is being denied the benefits of deployment of the advanced GM traits in the hybrids of genuine research-backed seed companies, as opposed to fly-by-night operators. The blame for this — widespread farmer demand-led, yet unregulated adoption of advanced GM technology — lies solely with the government. By blocking commercialisation and field trials of GM crops, based more on ideological positions than empirical regulatory science, it has turned India from a leading producer and exporter to a net importer of cotton.

The internet in a sachet — this idea's time has come



R S SHARMA

INDIA HAS done what no other country has: Made digital payments free, instant, and universal, and leads the world in mobile-data use. Yet, in one adjacent field we remain a laggard. Public Wi-Fi — the cheap, shared, last-mile connectivity that lets a citizen buy a little internet wherever she is — has barely taken root. India has around 4 lakh public hotspots. France, with a population a fraction of ours, has more than 13 million; the world has over 600 million. For a nation that prides itself on its digital public infrastructure, this is a gap worth explaining — and closing.

The idea was born in April 2016, in an email from Nandan Nilekani. Any consumer, he wrote, should be able to walk up to any Wi-Fi hotspot and connect with a single click — "like an Uber for hotspots" — and anyone with a spare connection could sell a little of it to a passer-by. At TRAI, we built exactly that: An open, interoperable architecture with no central gatekeeper, in which anyone could set up or aggregate a hotspot, and a single app would discover the nearest one, verify the user, take his few rupees through any wallet, and connect him seamlessly across every provider.

We called the retail outlet a Public Data Office, or PDO, and the analogy we kept returning to was the humble PCO — the public call office that put a telephone within walking distance of nearly every Indian by turning a shopkeeper into its keeper. The PDO was the PCO reborn for the internet: The corner shop or tea stall selling connectivity by the sachet, a rupee's worth at a time. That is why I have always thought of it as the internet in a sachet — for the genius of the Indian market is that the poor do not buy the bottle; they buy the sachet.

TRAI issued a consultation paper in July 2016, submitted formal recommendations in March 2017, and piloted the idea with eight entities running some six hundred hotspots. The pilot worked: One-time registration, interoperability, easy payment — all proven. The Prime Minister himself was shown the system in June 2018 and was greatly taken with it; he wanted it launched within weeks. Yet it was not until December 2020 that the framework was finally approved by the Union cabinet, as PM-WANI — the Prime Minister's Wi-Fi Access Network Interface.

Five years on, the promise remains largely unfulfilled. Why? The single biggest reason is the cost of backhaul: A small PDO is made to pay more for the very same fibre broadband that a household buys cheaply, which destroys the shopkeeper's economics before he has sold a single sachet. Add to this a persistent official tenderness towards incumbent telecom operators, never enthusiastic about a million small competitors selling connectivity outside their control.

The commonest objection now is that mobile data is so cheap that nobody needs Wi-Fi by the sachet. It is plausible, and wrong. Data now comes bundled in fixed monthly plans priced from a couple of hundred rupees, which the poorest cannot always afford, and which are of little help to a student joining an online class when the family's one data plan is on someone else's phone. The sachet has not become unnecessary; it has merely been abolished by the operators' tariff design, even as demand has exploded — online education, telemedicine, government services — precisely where mobile signal serves worst.

> What it requires is resolve — the government's willingness to back a consumer-first, open network against the pull of powerful incumbents

There is a strategic case too. Public Wi-Fi runs on unlicensed spectrum, sparing congested mobile networks the costly spectrum they buy at auction. Leaving a billion-plus people's connectivity entirely to two or three mobile companies, besides, is a dangerous dependence on a duopoly. A dense, open, independent Wi-Fi layer is a strategic de-risking of the nation's digital life — an alternative no single commercial interest can switch off.

What makes the delay especially galling is that the missing piece is already lying in the ground. BharatNet, begun in 2011 to carry optical fibre to every one of India's 2.5 lakh gram panchayats, has absorbed more than Rs 48,000 crore — more than five times what Aadhaar cost — and reached over 2 lakh of them. Yet it returns almost nothing, for a great trunk of bandwidth with no retail distribution is a pipe with no taps. Even a fully monetised BharatNet will leave gaps in hilly, forested terrain fibre may never reach — and that is where Starlink and the other satellite-broadband entrants see a market; Reliance Jio, too, could plausibly carry backhaul, subject to regulatory clearances still being worked through.

The two failures are each other's cure. BharatNet is a big pipe without a last mile; PM-WANI is a last mile starving for a cheap, big pipe. Converge them — let BharatNet act as a common carrier, selling bandwidth on fair, transparent terms to the small PDOs and aggregators, much as a power-transmission company carries electricity to its distributors — and you have, at a stroke, cheap rural broadband, a real alternative to the duopoly, and at last a use for the taxpayer's fibre. It needs two things the state has withheld: A capped, transparent wholesale price, and enforceable service guarantees. TRAI's June 2025 tariff order, capping what a PDO is charged for retail fibre, is a welcome first step that must be followed through.

None of this requires new invention: The architecture was sound from the first, the fibre is in the ground, and the need has only grown. What it requires is resolve — the government's willingness to back a consumer-first, open network against the pull of powerful incumbents. The day a student in a small town, or a labourer far from home, buys a few rupees of the internet from the shop on the corner and thinks nothing of it will have been worth the wait. It is time to bring the internet in a sachet to every Indian.

The writer is a former chairman of TRAI and former secretary, Government of India. Views are personal

India's science problem is not talent, but priorities



KAMINI WALIA

EVERY OCTOBER, the Nobel announcements trigger a familiar conversation in India: Why does India, with its enormous pool of scientific talent, rarely feature among them? Perhaps the more useful question is: What are the countries that produce transformative science doing differently and what holds our scientists back?

The answer is not simply money, though money matters. The US spends about 3.4 per cent of GDP on research and development, Germany over 3 per cent, Japan more than 3.5 per cent and South Korea over 5 per cent. China is approaching 2.7 per cent. India's gross expenditure on R&D is only about 0.64 per cent.

But the more revealing statistic is who funds research. In India, business contributes around 40 per cent of R&D expenditure, compared with roughly three-quarters in the US and China. Our research ecosystem therefore has both a scale and a private-sector participation gap. There is also increasing pressure on scientists to "develop technologies", "build products", "commercialise research" and "deliver solutions". These are legitimate national priorities but technology cannot be produced on demand. We are emphasising the last few stages while underinvesting in the first.

A diagnostic platform, for example, may require years of work on disease biology, pathogen behaviour, host response, biomarkers and molecular mechanisms before a product becomes possible. Basic science is the landscape from which innovation emerges. This is where India needs a rethink. First, government and research agencies should create a protected basic-science funding window. A defined proportion of competitive funding should support investigator-driven, curiosity-led research, without requiring every project to promise a patent, product or technology. Second, research funding should be explicitly divided into discovery, translation and deployment portfolios, with different timelines and success metrics. Third, India needs long-duration funding, with rigorous scientific review but considerably less administrative micromanagement. Fourth, we need a national shared research infrastructure. Genomics, proteomics, advanced imaging, biobanks, structural biology, AI and computational platforms should be accessible across institutions.

Finally, we must change how we see research and evaluate scientists. Publication counts, journal impact factors, patents and grant utilisation are convenient metrics, but they do not necessarily measure discovery. Scientific assessment should reward originality, reproducibility, important questions, new knowledge, datasets, platforms, technologies and influence on science or health policy. This means building the pipeline properly.

China offers an instructive example. As its overall R&D investment has grown, it has become a major global supplier of affordable technology. China produced around 80 per cent of the world's solar PV modules and battery cells in 2024, while its share of global electric-car production reached about 70 per cent in 2025. The lesson is to recognise the importance of connecting discovery to engineering, manufacturing and markets.

India has an extraordinary natural advantage. Our population provides a vast and diverse setting for research. The ambition should therefore be to create an ecosystem which attracts young scientists who can pursue an unconventional question for 10 or 15 years, fail, change direction and try again without being penalised for not producing an immediate product. If India wants the technologies of tomorrow, it must fund the science of today. The Nobel Prize should not be the target. Building a research system capable of producing Nobel-level discoveries should be.

> Basic science is the landscape from which innovation emerges. This is where India needs a rethink

The writer is senior scientist and programme officer, AMR, ICMR

WHAT THE OTHERS SAY

In a world turned upside down, Mr Trump blames Volodymyr Zelenskyy for failing to cut a peace deal. — *The Guardian*

READ. THINK. SHARE.
Scan the QR code to explore curated editorials and opinions from leading voices



On women's safety, move from reaction to prevention

MAYANK MAKHIJA
AND NAMAN SHARMA

THE 2012 Delhi gang-rape case was supposed to redefine women's safety in India. It led to unprecedented protests, sweeping legal reforms and the constitution of the Justice J S Verma Committee, which produced one of the most significant reports on criminal justice in independent India. Many of its recommendations were implemented through the Criminal Law (Amendment) Act, 2013. New offences such as stalking and voyeurism were introduced, the definition of rape was expanded and punishments became more stringent. Yet the Committee had cautioned that legal reform without institutional reform would remain incomplete.

Recent incidents in Delhi prove the warning was justified. A 17-year-old girl was allegedly gang-raped inside a sleeper bus that travelled nearly 47 km before she was abandoned. Days later, a 16-year-old girl was allegedly raped at knifepoint in a Delhi park by three men posing as police officers. Both incidents exposed grave failures in the safety of public spaces. The problem persists beyond individual offenders. Roads, parks, buses, metro stations, subways and footbridges cannot be permitted to become dangerous spaces due to preventable administrative failures. The debate is no longer confined to punishment; it is about the state's duty to make public spaces safe.

Modern criminology offers a useful explanation through routine activity theory. A crime becomes more likely when three conditions exist simultaneously: A motivated offender, a vulnerable target and the absence of effective guardianship. Poorly illuminated parks, non-functional CCTV cameras, unverified buses and inadequate police patrols reduce that guardianship. They do not create criminals, but they create opportunities for crime. According to the NCRB 2024, Delhi registered 1,058 rape cases with a rape rate of 14 per lakh women. More disturbing is the fact that these are only the reported cases; the countless crimes buried under fear, stigma and silence never find a place in official statistics. Media reports also indicate that at least 13 cases of sexual assault were reported in Delhi during September alone. The Supreme Court's decision to take *suo motu* cognisance reflects the seriousness of the institutional breakdown.

> The problem persists beyond individual offenders. Roads, parks, buses, metro stations, subways and footbridges cannot be permitted to become dangerous spaces because of preventable administrative failures

Why do severe punishments not prevent these crimes? The answer lies in the classical theory of penology developed by Cesare Beccaria. Beccaria argued that the certainty of punishment, rather than its severity, is the real deterrent. India today has stringent rape laws, including life imprisonment and capital punishment in specified circumstances. Yet offenders often believe they can evade immediate detection because surveillance is weak, policing is inadequate and investigations are delayed. A harsh law that is rarely enforced is a weaker deterrent than a moderate law that is enforced with certainty.

The conversation on women's safety must now move from reaction to prevention. First, every identified vulnerable public space in Delhi should undergo an independent safety audit covering lighting, CCTV functionality, emergency response systems and police visibility. Second, all commercial buses must have compulsory GPS tracking and live surveillance linked to transport authorities. Third, departmental accountability should accompany criminal prosecution whenever serious offences expose failures in surveillance, licensing or patrolling.

The Justice J S Verma Committee had identified "failure of governance" as the root cause of sexual crime. It recommended the creation of Rape Crisis Cells in every district, which would be immediately notified upon registration of an FIR and provide survivors with integrated legal aid, trauma counselling and psychological support from the first stage of investigation. While the government introduced One Stop Centres (Sakhi) in 2015, the Committee's vision of an automatic, police-linked rape crisis response has never been fully implemented.

Women should not have to calculate which road is safer, which park is brighter or whether a public bus can be trusted. The burden of safety cannot continue to rest on the victim's precautions instead of the state's preparedness.

Makhija is a Supreme Court lawyer and former special public prosecutor for the ED. Sharma is a final-year student at Maharashtra National Law University, Nagpur

A decade of placing research and innovation at the centre



ABHAY KARANDIKAR

ON JANUARY 3, 2019, at the Indian Science Congress, Prime Minister Narendra Modi gave the call for "*Jai jawan, Jai kisan, jai vigyan, jai anusandhan*". With these words, he articulated a vision placing research and innovation as the primary agenda for economic growth, strategic security, industrial development, entrepreneurship, and citizen-centric service delivery.

The facts and figures since 2014 reflect the breadth of this transformation. India rose from 81st in the Global Innovation Index in 2015 to 38th in 2025. It ranks third globally in the volume of scientific and engineering publications. The Gross Expenditure on Research and Development (GERD) has more than tripled from Rs 79,355.89 crore in 2013-14 to Rs 2,44,767.81 crore in 2023-24, reaching about 0.84 per cent of GDP. The private sector's share of GERD increased from 32.2 to 45.2 per cent over this period.

During the last decade, a total transformation of the institutional architecture of India's research base has been initiated. The Anusandhan National Research Foundation funds new research across universities, colleges and R&D labs while encouraging collaboration amongst academia, industry and government.

The Research, Development and Innovation (RDI) Fund, on the other hand, addresses a more difficult problem — the "valley of death", where technologies need pilot-scale demonstration and productisation just as risk and capital requirements rise sharply. The Rs 1 lakh crore RDI Fund represents a significant step towards addressing this financing gap and providing the patient, risk-tolerant capital that cutting-edge innovation requires.

In his 2023 Independence Day address, PM Modi recalled that when asked during a foreign visit whether Indian girls pursued science and engineering, he responded: "Today in my country more daughters than boys are taking up STEM." Initiatives such as Women in Science and Engineering (WISE)–KIRAN, Gender Advancement for Transforming Institutions (GATI) and Vigyan Jyoti have sought to create greater opportunities for women across the scientific and technological ecosystem.

Another pivotal step has been the transition towards building mission-mode capabilities around specific sectoral priorities for the country. The National Quantum Mission spans the full quantum stack, ranging from computing and communication to sensing, metrology, materials and devices, and supports research groups and start-up spin-offs. The IndiaAI Mission applies the same ecosystem logic to artificial intelligence, covering compute, datasets, skilling, foundational models and responsible AI. The Indian Semiconductor Mission has approved 10 projects across six states, including two fabs and eight packaging units, with Semicon 2.0 now pushing further into equipment and materials, indigenous IP, industry-led R&D and talent development.

Biotechnology is following a similar trajectory. The BioE3 Policy provides a broad foundation spanning bio-based chemicals and enzymes to precision biotherapeutics and space biotech. The bio-economy grew from about $10 billion in 2014 to roughly $195.3 billion in 2025 and is projected to reach $300 billion by 2030. The Deep Ocean Mission and Mission Mausam extend the integrated-ecosystem approach into resource and climate security.

The Indian Space Policy, 2023 and the establishment of IN-SPACe have opened the space sector to private participation across launch services, satellite manufacturing and downstream applications, with more than 400 space start-ups now operating in the ecosystem. Nuclear energy has also seen an opening, with the SHANTI Act providing a legal framework for private participation, subject to licensing, safety and regulatory requirements.

India's digital public infrastructure has been built over the decade around the JAM trinity of Jan Dhan accounts, Aadhaar, and mobile connectivity, and expanded through UPI, DigiLocker, Account Aggregator, and the Ayushman Bharat Digital Health Mission. In agriculture, satellites, drones, remote sensing, GIS, and AI are increasingly used to monitor crops, enable precision farming, estimate yields, and provide farmer advisories.

India's start-up ecosystem has also expanded rapidly during the last few years. Startup India, launched in 2016, put entrepreneurship at the core of the national development agenda. India is now the world's third-largest start-up ecosystem, with more than 2 lakh registered start-ups and over 130 unicorns. A wide range of schemes, such as the Seed Fund, Fund of Funds, Credit Guarantee Scheme, Atal Innovation Mission, NIDHI, BioNEST, iDEX, etc. launched since 2016 have strengthened access to incubation, mentoring, finance and markets. Under Modi 3.0, the focus has now sharpened towards emerging technologies.

As PM Modi stated, "The arms of science, technology and innovation must reach the poorest, the remotest and the most vulnerable person." The vision of science serving society lays a stronger foundation for India's aspirations, and as India moves towards Viksit Bharat 2047, the spirit of "*Jai vigyan, jai anusandhan*" captures this continuing commitment to scientific progress and innovation.

The writer is member, NITI Aayog and former secretary, Department of Science & Technology, Government of India

INSTRUCTIONS AGREED AT LAST WEEK'S GST COUNCIL MEETING

States, Centre told to focus on data analytics, tech to widen GST base

Aanchal Magazine
New Delhi, October 12

DEEPER DATA analytics, leveraging technology, moving away from enforcement-led and intrusive actions, tracking taxpayers who do not pay tax in cash and using invoice matching are among the measures that central and state officers have been asked to focus on to deepen the tax base under the Goods and Services Tax (GST) regime.

Following a series of process reforms announced at the 57th meeting of the GST Council last week, tax officials have also been asked to undertake granular data analysis to identify opportunities to expand the tax base. This includes closely monitoring transactions in which retail sales may be misdeclared as business-to-business (B2B) transactions to claim input tax credit (ITC), and identifying registered GST taxpayers who report no cash tax payments and instead set off their entire GST liability against eligible ITC.

Several business-friendly measures announced at the Council meeting, including the introduction of a Rs 10,000 threshold for issuing showcause notices under GST, are also likely to be rolled out before April 1, 2027, the implementation date for other measures.

"It is very important to look at stabilising the GST system. And that's why rate rationalisation was undertaken last year and now the process reforms. The GST authorities have been instructed to use technology and data analytics to detect evasion and make efforts to deepen the tax base. Although we have a good number of taxpayers registered under GST, there is scope to widen the tax base and we have asked states to widen the base," a senior official said.

"Businesses that are registered but showing anomalies in their GST declarations can have their transactions studied, and they can be nudged accordingly to become part of the tax system," the official added.

The technology-led GST data analytics initiative is expected to follow the approach of the Income Tax Department's NUDGE (Non-intrusive Usage of Data to Guide and Enable) campaign,



Tax officials have been asked to monitor retail sales misdeclared as B2B transactions to claim input tax credit. REUTERS

which uses risk analytics, behavioural insights and information sharing through global networks such as the Common Reporting Standard (CRS) and the Foreign Account Tax Compliance Act (FATCA) framework to nudge taxpayers towards compliance.

GST authorities will also rely on the Invoice Management System (IMS) to match credit details between sellers and buyers. Under a proposed additional layer to prevent mismatch notices, buyers will be able to accept, reject or hold each invoice as it appears. Accepted invoices will flow into their returns.

"A seller who wants to reduce something he has reported does it where the buyer will see it, so the two sides stay aligned. Much else rests on this," the official said.

Linking the input and output ledgers with the summary return would help identify fake credit as it arises, the official added.

To enable this invoice-matching mechanism, GST authorities will also introduce electronic invoicing (e-invoicing) as a step before the matching process. E-invoicing for business-to-consumer (B2C) transactions will be introduced from April 1, 2027, the official said, adding that the GST Network will provide the necessary support system to states for the transition. Under the existing GST e-Invoice System, businesses are required to upload B2B invoices to a government-approved Invoice Registration Portal (IRP) for validation and the generation of a unique Invoice Reference Number (IRN) and a QR code.

GST authorities are also expecting smoother movement of goods following the Council's recent decision to allow vehicles to be stopped only by tax officers from the state of the supplier or recipient, and only on the basis of "specific intelligence" and due authorisation. The measure was initially proposed only for supplier states. However, following detailed discussions at the Council meeting, recipient states were also included at the request of some states, the official said.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

IMF raises India's FY27 growth forecast to 7%

Siddharth Upasani
New Delhi, October 12

THE INTERNATIONAL Monetary Fund (IMF) has raised its GDP growth forecast for India to 7% for 2026-27, up from 6.4% in July, on the back of "stronger-than-expected momentum, supported by resilient domestic demand and robust services activity". The sharp increase in the forecast for the current fiscal comes days after the Reserve Bank of India (RBI) did the same. The Indian central bank now expects India's GDP to expand by 7.1% this year.

"India stands out with upward revisions, reflecting sustained momentum and growing evidence of stronger growth potential supported by favourable demographic dynamics, a large and expanding domestic market, and continued structural reforms," the IMF said in an update to its flagship World Economic Outlook report.

GDP growth in 2027-28 is also seen at 7%, up from the July estimate of 6.7%. The IMF's forecasts assume that the average price of crude oil — across Brent, Dubai Fateh, and West Texas Intermediate — will be $95.74 per barrel in 2026, up 41% from 2025. In 2027, the oil price is assumed to decline around 9% to $87.39. The sharply higher growth forecast for India is in contrast to the global economy. In 2026, the IMF now sees the world's output growing by 3.1%, only slightly higher than 3% in July. Meanwhile, global growth in 2027 is seen unchanged at 3.4%.

According to the multilateral agency, the negative supply shock from the war in West Asia is making its way through the world economy. And while the private sector has seemingly adjusted well to the conflict and trade-related shocks, the "possibility of larger economic costs" is becoming "more palpable". This is because the conflict is dragging on and buffers are becoming narrower.

Countries that are now projected to grow at a significantly higher rate in 2026 than in July include Germany (1.3% versus 0.7%) and UK (1.3% versus 1%). In fact, advanced economies as a whole are seen growing 2.1% in 2026, up from 1.7% before.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

LAW

Jantar Mantar de-notified: What courts have said about designating protest sites

Amaal Sheikh
New Delhi, October 12

THE DELHI Police on Monday de-notified Jantar Mantar as a designated protest site, naming Ramlila Maidan and Burari Ground in north Delhi and CBD Ground in east Delhi as alternatives.

According to the official notification, the decision follows a recommendation by a committee constituted by the Ministry of Home Affairs (MHA). The move comes amid the Delhi police's heavy barricading and crackdown on protesters in central Delhi since October 10. Protesters have gathered at several locations, including outside areas where prohibitory orders have been imposed under Section 163 of the BNSS.

The designation of protest sites is intended to regulate demonstrations while addressing concerns over traffic, public safety, and residents' rights.

Who decides Delhi's protest sites?

Article 19 of the Indian Constitution guarantees the fundamental rights to freedom of speech and expression, and to assemble peaceably, subject to reasonable restrictions — including in the interests of "public order".

Unlike other states, Delhi's police and public order fall outside the powers of the Delhi Assembly. Under Article 239AA(3)(a), the Legislative Assembly has the power to make laws for NCT Delhi except on three issues: public order, police, and land.

The Delhi Police thus function under the MHA. The police regulate demonstrations, including applications for permission and compliance with conditions governing designated sites. Organisers had to apply 10 days in advance to protest at Jantar Mantar, which had a cap of 1,000 people. Ramlila Maidan was another designated site, with a capacity of around 50,000 people.

Law and order vs 'public order'

While the police often cite law and order when regulating protests, Article 19 permits restrictions on fundamental rights on specified grounds, including "public order". The Supreme Court has treated them differently.

In the 1966 SC ruling of *Ram Manohar Lohia v. State of Bihar*, the court observed that an act may affect law and order without affecting public order. It described law and order, public order, and security of the state as three concentric circles, with public order occupying a narrower category.

"Law and order represent the largest circle within which is the next circle representing public order and the smallest circle represents security of State. It is then easy to see that an act may affect law and order but not public order, just as an act may affect public order but not security of the State," it said.

In *Re: Ramlila Maidan Incident* (2012), the court said a restriction imposed to maintain law and order "would be least intruding into the guaranteed freedom", while public order "may qualify for a greater degree of restriction" since it was of greater social concern. The court clarified that "every breach of peace may not always lead to public disorder."

Speaking to *The Indian Express*, senior advocate Gopal Sankaranarayanan explained that law and order issues can involve localised disturbances, such as two people fighting on a street. Public order, he said, concerns disturbances that affect the wider community, such as large-scale disruption of transport.

"Law and order is not a valid basis to limit the right to speech and expression," he said, adding that the police cannot invoke public order on their own to justify restricting fundamental rights. "The first step is a law needs to be made which gives police that power. Then the police exercise the power under that law," he said, adding that the police must identify the law under which they are exercising their powers.

Professor G Mohan Gopal told *The Indian Express* that dissent is the "lifeblood of a peaceful and law-abiding society" and democracy exists to accommodate different opinions. He added that shutting down designated protest sites and suppressing peaceful dissent under "law and order" grounds risks driving citizens to desperation, "because all the legitimate channels, democratic channels for defending the public order based on democracy have been closed".

How can protest sites be designated?

The bench in *Himat Lal K. Shah v. Commissioner of Police, Ahmedabad* (1972), held that the police could require prior permission for public meetings on streets. But it added that the State "cannot by law abridge or take away the right of assembly by prohibiting assembly on every public street or public place".

In *Mazdoor Kisan Shakti Sangathan v. Union of India* (2018), the case pertained to restrictions on demonstrations at Jantar Mantar and moving them to Ramlila Maidan. The petitioners argued that the ground would be too far from the "concentration of 'power'" in New Delhi, and would impact the protests' goals, concerning the very authorities the slogans are raised against.

While the court accepted that contention, it underlined the need to balance the rights of protesters with those of residents and held that "while regulating such demonstrations in public interest, particular areas can be earmarked". It thus allowed a limited portion of Jantar Mantar Road to remain open for regulated demonstrations, noting that the site had been used for protests for a considerable period.

NOBEL PRIZE IN ECONOMICS 2026

How to keep an economy stable using policy tools

The Great Depression prompted policymakers to advocate government intervention in economic crises. Hall and Woodford's work refines that idea



UDIT MISRA

THE ROYAL Swedish Academy of Sciences has awarded the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2026 — commonly called the Economics Nobel — to two American economists: Robert E Hall of Stanford University and Michael Woodford of Columbia University. They will share the prize money of 12 million Swedish kronor (approximately Rs 12 crore).

What have they been awarded for?

The official citation states they have received the award "for microeconomic foundations of macroeconomic stabilisation". Macroeconomic stabilisation refers to policies that governments and central banks use to "stabilise" an economy.

For instance, the Reserve Bank of India last week raised interest rates with a view to contain inflation. Inflation in India has been going up owing to a rise in crude oil prices and prices of other commodities in the wake of the US-Iran war, the blockade of the Strait of Hormuz, etc. There are also worries that inflation will rise further because deficient rainfall and higher temperatures due to El Niño may hurt food production and push up food prices.

If the government and the central bank followed the dictum that the government should not intervene and that the markets will price things correctly on their own, inflation can potentially soar quite high. That would hurt most Indians, since they are not rich enough to handle such a situation. It could lead to people cutting down all other expenses to focus on the essentials, which would cause many businesses to fail and lots of job losses. The economy, as a whole, may take a much longer time to recover.

But an RBI intervention could stabilise the path in such a manner that the situation never becomes as dire as it could become if nothing was done.

Similarly, over the past few years, the Union government has been cutting taxes — both direct taxes (personal income tax and corporate tax) and indirect taxes (GST) — with a view to boost consumption by individuals and investments by companies. That's because in the years leading up to and during the Covid-19 pandemic (and resulting lockdowns), India's economy struggled.

Through tax cuts, the government tried to boost economic activity — the RBI also helped in those years by cutting interest rates. Without such interventions, a recovery might have taken much longer.

These are the examples of macroeconomic "stabilisation" policies that academics such as Hall and Woodford helped refine through their work.

To understand the "microeconomic foundations" of these macro policies, one has to understand the trigger for their work.

What is their contribution?

A crash in the stock markets in October 1929 marked the start of the Great Depression in the US. As the days and months rolled by, the US gross domestic product — the value of annual output — fell by 30% and the unemployment rate soared to 25%.

Thousands of banks and businesses went under. The crisis reverberated through the world, especially affecting Europe.

'For microeconomic foundations of macroeconomic stabilisation'



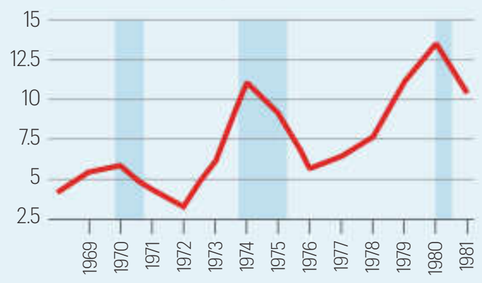
Developed new macroeconomic models, leading to policies that reduced the economy's peaks and troughs.



ROBERT E HALL, 83
Stanford University, U.S.



MICHAEL WOODFORD, 71
Columbia University, U.S.



SOURCE: WORLD BANK



SHADED AREAS INDICATE U.S. RECESSIONS
SOURCE: U.S. BUREAU OF LABOR STATISTICS

Demand and supply

Hall focused on the demand side of the economy: how do people decide how much they would consume, and when and how do they choose to raise or cut consumption.

Woodford's work focused on the supply side: how firms behaved and took decisions such as changing prices.

The dominant policy paradigm leading up to the Great Depression was that governments should not intervene, letting markets resolve through forces of demand and supply instead. But the widespread devastation made everyone rethink.

British economist John Maynard Keynes provided a solution that seemed to work: Governments and central banks should "stabilise" the economy through policy interventions. When growth is slow (and unemployment high), policymakers should cut interest rates and taxes, and the governments should spend more even at the cost of borrowing money itself and having a budget deficit, thus incentivising economic activity.

When the economy starts overheating and the inflation rate starts rising fast, the policymakers should slow down the economy by policies like raising interest rates. Thus, government intervention could save economies from suffering from wild fluctuations.

This broad policy formula worked until the experience of the 1970s. As the *charts* show, this was a decade when economies faced high inflation and high unemployment both at the same time, thanks to repeated oil price shocks. Efforts to boost a slowing economy only spiked inflation. Efforts to contain inflation further worsened unemployment. The situation became so bad (shaded areas in the *charts* show recessions) that the Keynesian recipe of government intervention was seen as the problem, not the solution.

Through their work from the late 1970s onwards, Hall and Woodford revitalised the Keynesian approach by making it grounded in research. Keynes had assumed a linear or straight relationship between variables such as inflation and unemployment, and consumption and production. Hall and Woodford started studying the ground-level data.

Hall focused on the demand side of the economy: questions such as, how do people decide how much they would consume, and when and how do they choose to raise or cut consumption. A better understanding of these, based on household data, has informed policy choices.

For instance, in 1982, he used microdata for a number of individual households to arrive at the insight that not everyone in the economy increases consumption based on income and future expectations; households that are living hand-to-mouth do not follow that assumption.

Woodford's work focused on the supply side: how firms behaved and took decisions such as changing prices. The assumption is that markets find the ideal price based on supply and demand, but businesses don't change prices everyday. He found that businesses look to the future when fixing prices. This means that expectations about future interest rates are important for inflation. In turn, this had a tremendous policy implication: Central banks must clearly convey what they intend to do, which is important for firms when they fix prices.

What is its significance?

Over the past few years, the global economy has faced an exceptional number of risks and challenges: from a global pandemic that brought all economic activity to a sudden halt, to a growing number of wars that have spiked prices of essential commodities, to policy missteps like tariffs.

And yet, far from a recession, the economic growth continues to surprise on the upside. A big reason why policymakers worldwide have been able to avoid worse outcomes is because their policies are informed by the work of academics such as Hall and Woodford.

CULTURE

Garba in Gujarat: From dancing in villages to multi-crore enterprise

Ritu Sharma
Ahmedabad, October 12

THE NINE nights of the Navratri festival, marked by dancing the Garba to welcome the goddess Ambe Maa, began in Gujarat on Sunday.

Navratri and Garba

Garba, inscribed on the Representative List of the Intangible Cultural Heritage of Humanity by UNESCO in December 2023, is the devotional ritual dance once performed only by women in their homes and in temple courtyards during Navratri. Over the past few decades, Garba has crossed geographical and cultural boundaries.

In Gujarat itself, Navratri is a multi-crore economic phenomenon that has been taking new forms every season. There are theme-based Navratri nights; there has been a resurgence of the traditional form of 'mandli' and 'sheri' Garba, "sustainable and eco- friendly" Garbas, as well as events dedicated to pet animals. The dancing happens in venues from open grounds to humongous air-conditioned domes.

While Garba can be performed at any celebration or happy occasion, the Navratri Garba is Gujarat's true community celebration. Social groups, *Yuvak Mandals*, apartment complexes, event planners, and clubs in cities all organise Garba nights. The accompanying musicians can range from a single *dholi* (drummer) to large ensembles.



Garba rehearsals in Ahmedabad on Saturday night. EXPRESS

How it started, evolved

Prof Parul Shah, former dean and head (dance), at the Faculty of Performing Arts, Maharaja Sayajirao University of Baroda in Vadodara, said the word '*Garbo*' or '*Garba*' comes from the Sanskrit word '*Garbhdeep*', which refers to an earthen pot with circular holes, and is popularly known as '*Garbo*'.

"There isn't a clear date for the beginning of Garba, but the celebration is associated with both the dynasties that ruled Gujarat and with the peasantry and agricultural classes. Singing and dancing with the flute and other musical instruments as part of dance forms like Garba, Garbi, and Raas, and variations such as dandia or the 'stick raas', is believed to date back to the age of Lord Krishna," Shah said.

Living heritage

Garba was inscribed on the Representative List of the Intangible Cultural Heritage of Humanity by UNESCO in December 2023.

"Then, with the establishment of the Ambaji Shaktipeeth around 700 years ago, people started worshipping the goddess through the Garba. The earthen pot is a symbol of the human body, and the lamp inside signifies the divine soul. The festival is dedicated to the worship of the feminine energy or *Shakti*," Shah, who was among the experts who prepared the dossier for the nomination for the UNESCO designation, said. "Women would do the *sthapana* (installation) of Mataji in the chowk of their village temple, do *aarti*, and dance the Garba," Shah said.

The antiquity of Garba songs is difficult to establish precisely, Shah said. In the 20th century, Garbas were mainly passed down through generations of women in families and communities, she said. The artisans who craft Garba objects also pass on their skills orally and through practice.

Ghazal singer Atul Brahmbhatt, whose father Jitu Bhagat started the Sri Jitu Bhagat Satsang Mandali in Ahmedabad in 1949, told *The Indian Express* that his father was initially criticised for participating in an activity that was traditionally the preserve of women.

Atul said Jitu was influenced by the Vallabh and Dhola brothers who are credited with having written the 118-paragraph '*Anand no Garbo*', sung even today. "My father wrote hundreds of Garbas. Since there was no recording, devotees would memorise them and go back to their villages to sing," Atul said.

Changing Garba events

"Garba has been redefined by every generation through new performative styles, contemporary lyrics, and changing fashion trends, while keeping the circle of feminine energy or *Shakti* intact. It remains a living tradition," Prof Shah said.

However, some old-timers who see Garba as essentially an expression of devotion are put off by the modern commercial dimensions the celebration has acquired.

Then there is the increasingly visible presence of communal elements who have in recent years sought to associate themselves with Navratri.

A multi-crore celebration

The Navratri Garba celebrations are a massive economic enterprise — from the owners of party plots, planners of the event, choreographers, and providers of sound and light systems, to those involved in the food industry, and the artisanal community of potters, blacksmiths, and carpenters. Then there is the cloth and jewellery industry.

Many schools and universities offer professional courses in dance and music, costume and ornament design, landscape planning, and sound and light design, which contribute to Garba creations.